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GIGA Display Shield Touch Interface

Learn how to use the touch interface
on the GIGA Display Shield.

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The **GIGA Display Shield** has an
advanced touch interface,
supported via the
Arduino_GigaDisplayTouch
library.

This library is used to return the
number of contact points, and the
`x, y` coordinates for each of
contacts, and in this guide you will
learn about different methods to
obtain the data.

Hardware & Software Needed

Arduino GIGA R1 WiFi

**Arduino GIGA Display
Shield**

Arduino IDE

Overview

The **Arduino_GigaDisplayTouch**
library can be used in
combination with any of the

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- Delay Example
- GFX Touch Example

available options), but is independent from them and works standalone.

The library has two methods of reading sensor data:

Polling - continuously read the sensor data through the `getTouchPoint()` method.

IRQ - only read data when the display is touched through an interrupt and a callback function, using the `onDetect()` method.

The number of contacts (fingers touching) can be retrieved and stored as an integer:

```
1 contacts = touchDetecto
```

To read the `x` and `y` coordinates, we need to use a specific `struct` called `GDTpoint_t` which contains the values. The example belows simply iterates through the number of `contacts` and stores the coordinates of each contact.

```
1 GDTpoint_t points[5];
2 uint8_t contacts;
3
4 for (uint8_t i = 0; i <
5     x_coordinate = points
6     y_coordinate = points
7 }
```

Touch Interface

Gesture Detection

As you are able to retrieve multiple touch points at very fast rates, it is possible to create **gesture detections**. You could for example:

- change layout when swiping three fingers right,
- zoom in and out using two fingers,
- adjust the volume by swiping a finger from the bottom right corner and up,
- and many more scenarios.

This library can together with the supported graphics libraries create really interactions that we are using in modern day smartphones, tablets etc., with an easy interface and very fast response times.

Polling Example

The polling example demonstrates how to continuously read the touch sensor using the `getTouchPoints()` method. Whenever the display screen is touched, the number of contacts + the coordinates of each contact is printed to the Serial Monitor.



Code Source on [Github](#)

```
1  /*
2   * Touch_Polling.ino
3   *
4   * This example shows
5   * first touch points
6   *
7   * The setup() functi
8   * controller. The lo
9   * is detected. Whene
10  * points and first t
11  *
12  * For the interrupt
13  *
14  * Instructions:
15  * 1. Connect your GI
16  * board .
17  * 2. Upload this ske
18  * 3. Open the Serial
19  * 4. Touch the scree
20  * the Serial Monitor
21  *
22  * Initial author: Le
23  * Created: 03 May 20
24  */
25
26 #include "Arduino_Gig
27
28 Arduino_GigaDisplayTo
```

IRQ Example

The IRQ example demonstrates how to set up an interrupt that triggers a function anytime the screen is touched. The interrupt is set up inside of the `setup()` function, using the `onDetect(function)` method.

 Code Source on [Github](#)

```
1  /*
2   * Touch_IRQ.ino
3   *
4   * This example shows
5   * first touch points
6   *
7   * The setup() functi
8   * controller. Whenev
9   * is called that pri
10  * the Serial Monitor
11  * is configured to w
12  *
13  * For the polling ve
14  *
15  * Instructions:
16  * 1. Connect your GI
17  * board .
18  * 2. Upload this ske
19  * 3. Open the Serial
20  * 4. Touch the scree
21  * the Serial Monitor
22  *
23  * Initial author: Le
24  * Created: 03 May 20
25  */
26
27 #include "Arduino_Gig
28
```

Delay Example

An important factor to consider is that the `loop()` on the GIGA R1 WiFi is executed at a very fast rate, meaning that you will register several touches each time you touch the screen.

This means that whenever you tap the screen, even quickly, you register somewhere between **5-20 touches**. So if you want a specific function to only execute once on a specific touch point, you will need to implement a delay in your code.

Using the conventional but

method is possible but not ideal.
The best method to register only a single touch is through using the `millis()` method.

The example below is based on the **Polling Example**, and limits the if statement to only execute once every `250` milliseconds. This can be edited in the `threshold` variable.

```

1  #include "Arduino_Gig
2
3  Arduino_GigaDisplayTo
4  int lastTouch;
5  int threshold = 250;
6
7  void setup() {
8      delay(3000);
9      Serial.begin(115200
10     while(!Serial) {}
11
12     if (touchDetector.b
13         Serial.print("Tou
14     } else {
15         Serial.print("Tou
16         while(1) ;
17     }
18 }
19
20 void loop() {
21     uint8_t contacts;
22     GDTpoint_t points[5
23
24     contacts = touchDet
25
26     // check if more ti
27     if (contacts > 0 &&
28         Serial.print("Con

```

GFX Touch Example

The below example requires uses the `Arduino_GigaDisplay_GFX` library, and demonstrates how to

touch the screen. It implements the `millis()` function to limit the number of executions.

Anytime the screen is touched, the background and text color inverts (black and white).

```
1  #include "Arduino_Gig
2  #include "Arduino_Gig
3
4  Arduino_GigaDisplayTo
5  GigaDisplay_GFX displ
6
7  #define WHITE 0xffff
8  #define BLACK 0x0000
9
10 #define screen_size_x
11 #define screen_size_y
12
13 int touch_x;
14 int touch_y;
15
16 int lastTouch;
17 int threshold = 250;
18
19 bool switch_1;
20
21 void setup() {
22     delay(3000);
23     // put your setup c
24     Serial.begin(9600);
25     display.begin();
26
27     if (touchDetector.b
28         Serial.print("Tou
```

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